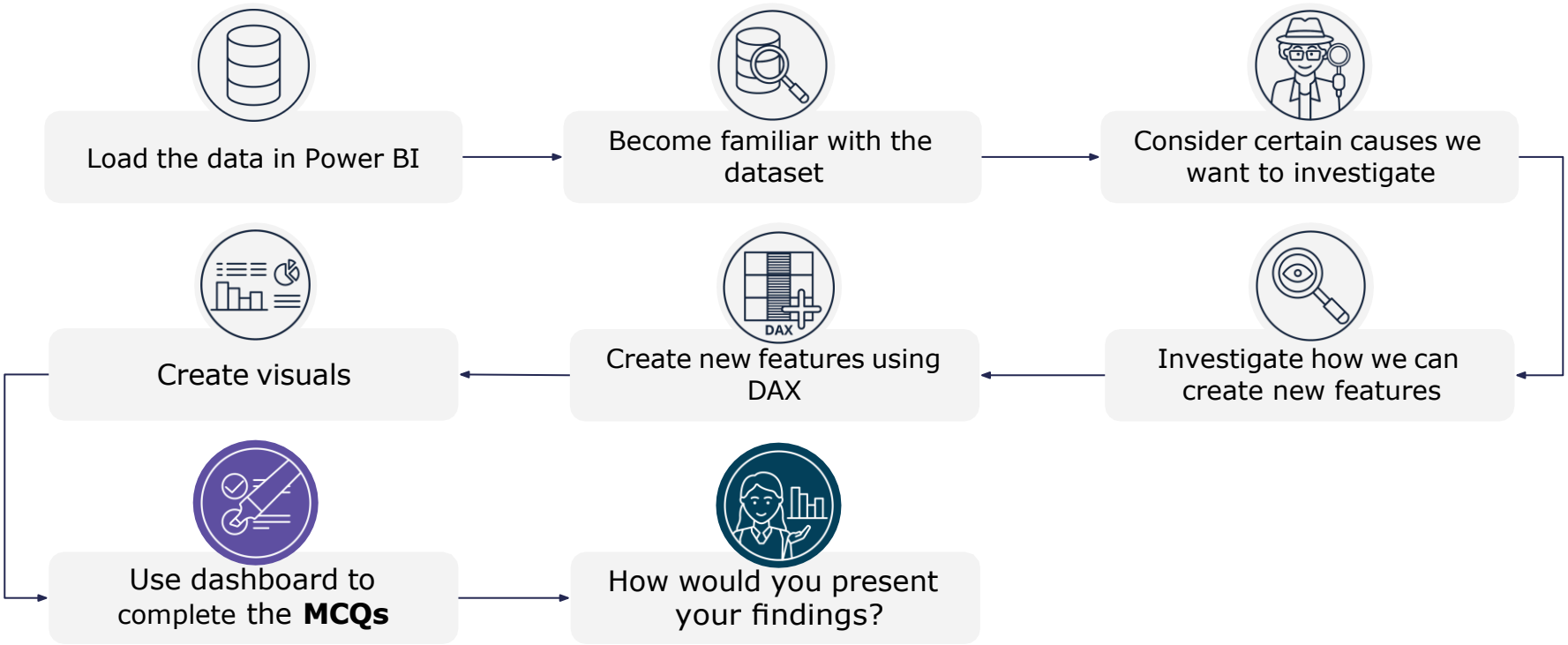
Case Study Walk-through : Human Resources (Overview)

HR Case Study Instructions

You are a manager or **analyst** at a medium-sized company. You have noticed a **drop in performance** from the team. You **investigate the causes** and find that **absenteeism** is quite a problem. You need to determine how to help those who are missing work.

You are provided with a **Human Resources Absenteeism** dataset. You are required to use this dataset and build a **Power BI dashboard**. This dashboard will then be used to **gather insights** and **answer questions** regarding employee absenteeism.

Instructions overview



Complete the following steps

1. Make sure that you have the Power BI Desktop installed on your device or virtual machine.
2. Download the **Human Resources Absenteeism** dataset (two CSVs) and open in Power BI.
3. Watch (or attend) Case Study Walk-through: Human Resources (**Problem Statement**).
4. Write down any additional features that may describe absenteeism?
5. Watch (or attend) Case Study Walk-through: Human Resources (**Features and DAX**).

Complete the features

6 Complete the following features:

In the **Absenteeism** table:

- **Month** based on *Month of absence;*
- **Season** based on *Month of absence*;
- **Absenteeism reason** based on *Reason for absence;*
- **Disciplinary status** based on *Disciplinary failure*; and
- **Week day** based on *Day of the Week*.

In the **Employee_info** table:

- **BMI** based on *Weight* and *Height*;
- **Drinker status** based on *Social drinker*;
- **Smoker status** based on Social smoker; and
- **Education level** based on Education.

7 Watch (or attend) Case Study Walk-through: Human Resources (**Visuals in Power BI**).

8 Use your Power BI dashboard to complete the MCQs.

Do not wait for the walk-throughs to attempt these MCQs!

9 Think about how you would present your findings.

Become familiar with the dataset

Let's look at the **descriptions** we have available on each feature (column) of the dataset.

Employee_info *table*

ID	Unique identifier for each employee.
Transportation expense	Transportation cost (in USD).
Distance from residence to work	The distance from the place of work and the employee's place of residence (kilometers).
Age	Age of the employee (years).
Education	High school (1), graduate (2), postgraduate (3), master and doctor (4).
Children	Number of child dependents.

Become familiar with the dataset

Let's look at the **descriptions** we have available on each feature (column) of the dataset.

Employee_info *table*

Social drinker	Either "yes" (1) or "no" (0).
Social smoker	Either "yes" (1) or "no" (0).
Pet	The number of pets an employee has.
Weight	The weight of the employee (in kilograms).
Height	The height of the employee (in centimeters).

The employee information table has 11 (original) features and 37 rows.

Become familiar with the dataset

Let's look at the **descriptions** we have available on each feature (column) of the dataset.

Absenteeisms *table*

ID	Unique identifier for each employee.
Reason for absence	Code 0 : Incomplete submission Code 1, 2, 3 and 4 : Family-related Code 5 - 25, 27, 28 : Medical reasons Code 26 : Unjustified leave
Month of absence	Month number of the year.
Day of the week	Monday (2), Tuesday (3), Wednesday (4), Thursday (5), Friday (6).
Work load Average/day	Average number of work tasks completed per day.
Hit target	Target hit factor (0-100).

Become familiar with the dataset

Let's look at the **descriptions** we have available on each feature (column) of the dataset.

Absenteeisms *table*

Disciplinary failure	A failure to adhere to a previous disciplinary warning relating to absenteeism (yes=1, no=0).
Absenteeism time in hours	Number of hours the employee has been absent.
Service time	Number of working hours produced by each employee.

The absenteeisms table has 9 (original) features and 639 rows.

Spring: March, April, and May
Summer: June, July, and August
Autumn: September, October, and November
Winter: December, January, and February

Complete the MCQs

1	Which season had the most absenteeisms reported?
2	Which month had the highest number of absenteeisms logged in the second-highest season, and how many absenteeisms were reported?
3	There is a month 0 that was reported three times for absenteeism. What was the reason for these absenteeisms?
4	How many family-related absenteeisms were reported?
5	Which code was the highest reported reason for family-related absenteeism, and how many were logged for this code?
6	How many employees are classified as overweight and obese?
7	Which employee matches the following criteria? - Overweight (not obese) - Social drinker - Social smoker

Complete the MCQs

8	How many times did the employee from question 7 report absenteeism, and what were the reasons?
9	What percentage of the absences reported from the employee in question 7 were in the summer?
10	How many employees have at least two children and one or more pets?
11	**True or false?** Employees in their 20s were absent more times on Mondays and Fridays than on other days.
12	How many employees disregarded disciplinary warnings, and what was the reason for their absence?
13	What percentage of the employees from question 12 had a high school education level?
14	**True or false?** Based on the data, all the employees with a high school education level and a hit target value of less than 90 disregarded disciplinary warnings.
15	Which season had the highest work load average per day?
16	How many employees were only absent from work once, and what similarities do they share?